SETTLING AND SEDIMENTATION

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Filtration versus settling and sedimentation

Filtration-force acting on fluid and require a filter medium

Sedimentation-force acting on particles and does not require filter medium

- **Filtration versus settling and sedimentation:**

Filtration

☞ The solid particles are removed from the slurry by forcing the fluid through a filter medium, which blocks the passage of the solid particles and allows the filtrate to pass through.

Settling and sedimentation

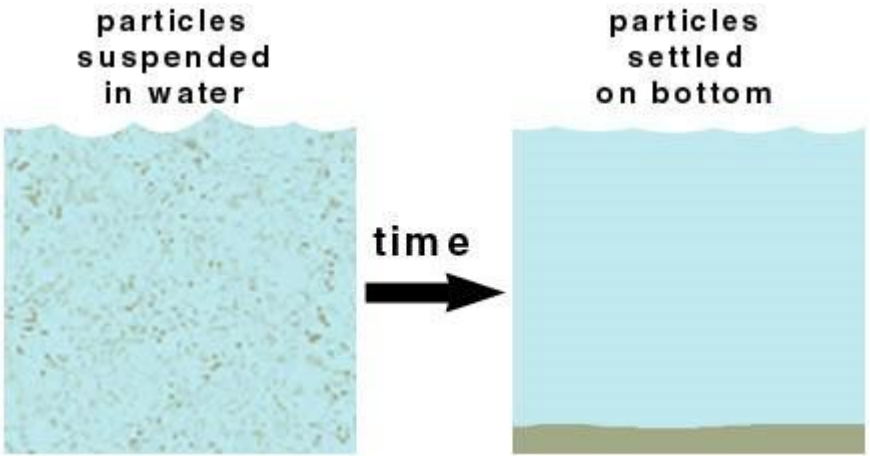
☞ The particles are separated from the fluid by forces acting on the particles.

What is sedimentation??

- Sedimentation is the process of allowing particles in a suspension of water to settle out of the suspension under the effect of gravity.

- Sedimentation describes the motion of molecules in solutions or particles in suspensions in response to an external force such as gravity, centrifugal force, or electric force.

- The separation of a dilute slurry or suspension by gravity settling into a clear fluid and slurry of higher solids content is called sedimentation.



Objective & Applications

- **Objective**
  - The purpose is to remove the particles from the fluid stream so that the fluid is free of particle contaminants.

- **Applications**
  - Removal of solids from liquid sewage wastes
  - Settling of crystals from the mother liquor
  - Separation of liquid-liquid mixture from a solvent-extraction stage in settler
  - Water treatment
  - Separation of flocculated particles
  - Lime-soda softening iron and manganese removal
  - Wastewater treatment

Free settling versus Hindered settling

Free Settling	Hindered settling
Falling of a particle is not affected by other particles and wall container/column.	When particles are too crowded, a particle's motion is impeded by other particles present in the container
A particle is at a sufficient distance from the walls of the container and from other particles so that the fall is not affected.	Exists when particles are very near each other even if there is no actual collision
Its fall is not affected by other particles	Particles settle at a lower rate
Free settling velocity is the terminal velocity (vt)	

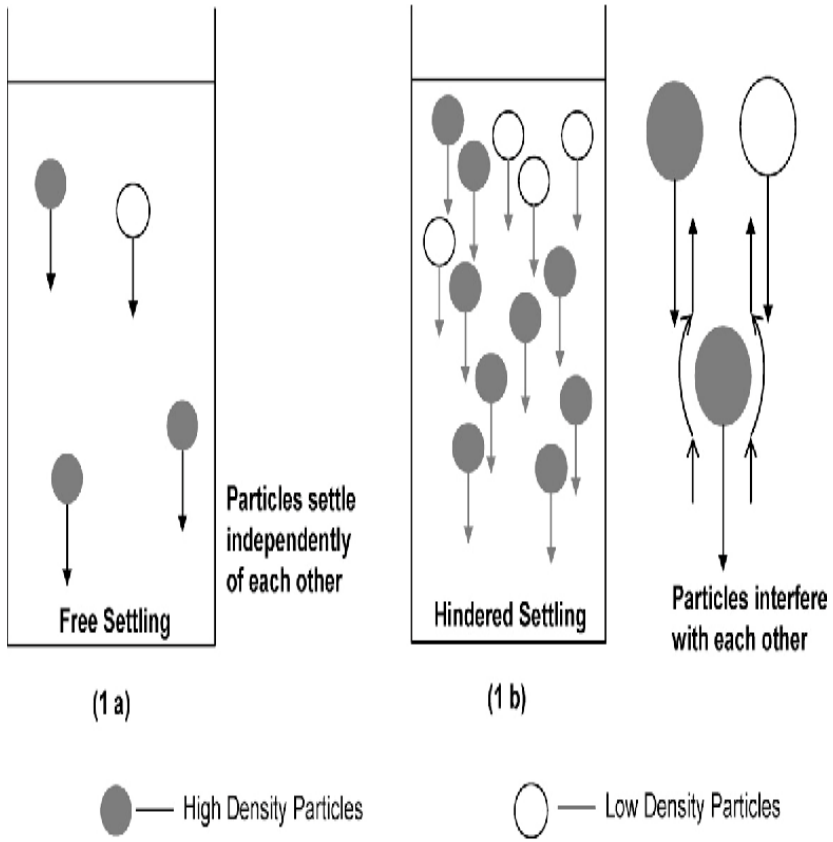


Fig. 1. Schematic representation of free settling and hindered settling.

Theory for sedimentation

- Whenever a particle is moving through a fluid, a number of forces will be acting on the particle.
- First, a density difference is needed between the particle and the fluid.
- If the densities of the fluid and particle are equal, the buoyant force on the particle will counterbalance the external force and the particle will not move relative to the fluid.
- There are three forces acting on the body:

- Gravity Force
- Buoyant Force
- Drag Force

THEORY OF PARTICLE MOVEMENT THROUGH A FLUID

For a rigid particle of mass m moving in a fluid, there are three forces acting on the body:

(1)Gravity force, F_g, acting downward

$$F_g = mg$$

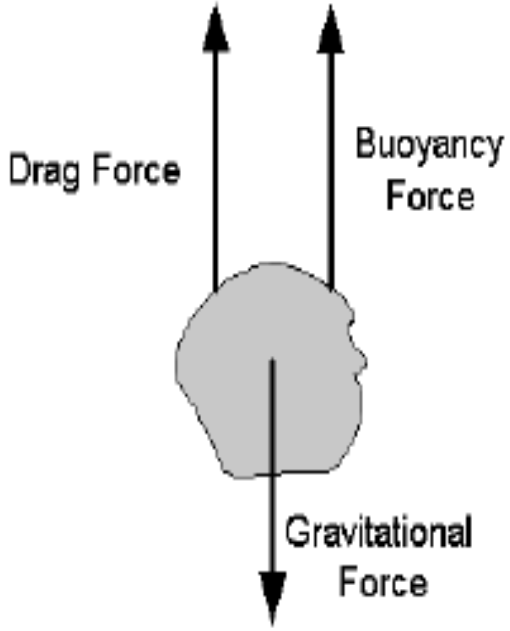
(2) Buoyant force, F_b, acting upward

$$F_b = \rho g V_s = \frac{m\rho g}{\rho_s}$$

where ρ = density of the fluid

ρ_s = density of the solid particle

V_s = volume of the particle



(3) Drag force, F_D, acting in opposite direction to the particle motion

$$F_D = C_D A \frac{\rho v^2}{2}$$

where C_D = the drag coefficient

A = the projected area of the particle

The resultant force equals the force due to acceleration.

$$m\frac{dv}{dt} = mg - \frac{m\rho g}{\rho_s} - C_D A \frac{\rho v^2}{2}$$

This comes directly from Newton's Second Law:

$$\sum F = ma = m\frac{dv}{dt}$$

The falling of the body consists of two periods:

(1) The period of accelerated fall

- ☞ **The initial acceleration period is usually very short, of the order of a tenth of a second or so.**

(2) The period of constant velocity fall

Set $\frac{dv}{dt} = 0$ **and solve the above equation for *v*.**

$$m\frac{dv}{dt} = mg - \frac{m\rho g}{\rho_s} - C_D A\frac{\rho v^2}{2} = 0$$

$$\square \qquad v = v_g = \sqrt{\frac{2(\rho_s - \rho)mg}{\rho_s \rho C_D A}}$$

* v_g **is called the free settling velocity or terminal velocity.**

$$v = v_g = \sqrt{\frac{2(\rho_s - \rho)mg}{\rho_s \rho C_D A}}$$

For spherical particles of diameter *d*,

Density= mass/volume
Mass=density x volume

$$m = \frac{\rho_s \pi d^3}{6} \text{ and } A = \frac{\pi d^2}{4}$$

$$\square \quad v_g = \sqrt{\frac{4(\rho_s - \rho)dg}{3\rho C_D}}$$

- **Terminal settling velocity**

When a particle is dropped into a column of fluid it immediately accelerates to some velocity and continues falling through the fluid at that velocity

- The **speed of the terminal settling velocity** of a particle depends on the properties of both the fluid and the particle:

- Properties of the particle include:

✓ The size of the particle (d).
✓ The density of the material making up the particle
✓ The shape of the particle.

The drag coefficient for rigid spheres has been shown to be a function of the Reynolds number.

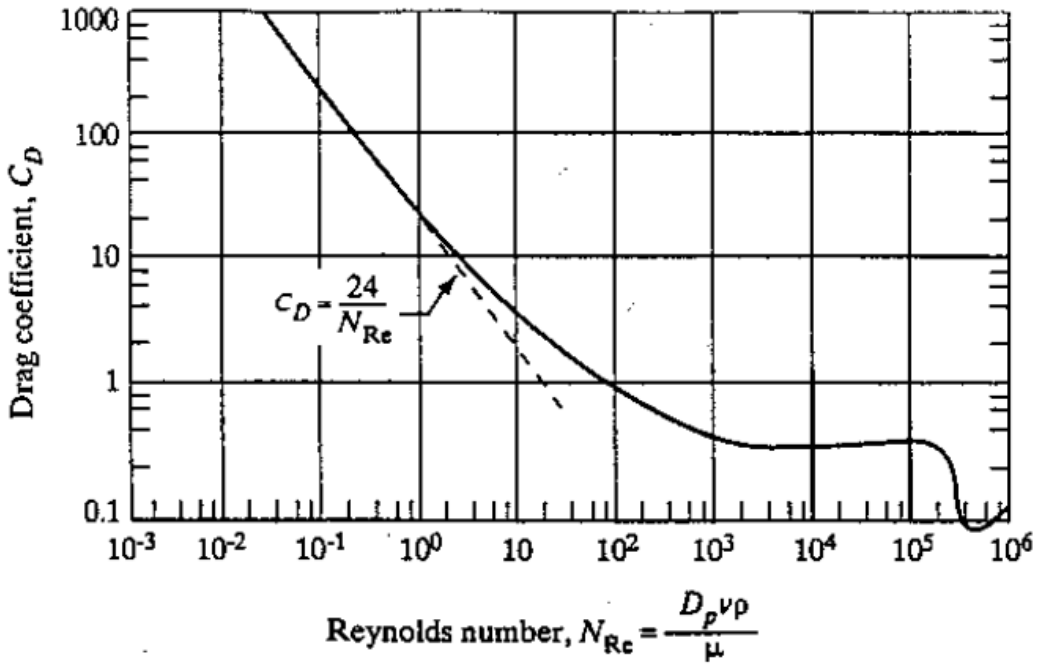


FIGURE 14.3-1. *Drag coefficient for a rigid sphere.*

Drag Coefficient

For particle movement in fluids, drag force is resistance to its motion.
The drag coefficient is a coefficient related to drag force.
The overall resistance of fluids act to particles can be described in term of drag force using drag coefficient.

Sphere object
At very low Reynolds number (<0.2), Stoke law is applicable.
The inertia forces may be neglected and those of viscosity alone considered.

In the Stokes' law region ($N_{Re} < 1$),

$$C_D = \frac{24}{N_{Re}} = \frac{24}{\rho v d / \mu}$$

$$v_g = \sqrt{\frac{4(\rho_s - \rho)dg}{3\rho C_D}} = \sqrt{\frac{4(\rho_s - \rho)dg}{3\rho} \times \frac{\rho v_g d}{24\mu}} = \sqrt{\frac{d^2}{18\mu}(\rho_s - \rho) g v_g}$$

$$\square \quad v_g = \frac{d^2}{18\mu}(\rho_s - \rho) g$$

Free settling velocity or terminal velocity or stoke's law settling

Terminal settling velocity is the constant velocity at which a particle falls through a fluid when the net (resultant) force on it becomes zero.

[Example] Many animal cells can be cultivated on the external surface of dextran beads. These cell-laden beads or "microcarriers" have a density of 1.02 g/cm³ and a diameter of 150 μm. A 50-liter stirred tank is used to cultivate cells grown on microcarriers to produce a viral vaccine. After growth, the stirring is stopped and the microcarriers are allowed to settle. The microcarrier-free fluid is then withdrawn to isolate the vaccine. The tank has a liquid height to diameter ratio of 1.5; the carrier-free fluid has a density of 1.00 g/cm³ and a viscosity of 1.1 cP.
(a) Estimate the settling time by assuming that these beads quickly reach their maximum terminal velocity. (b)

- Height/diameter= 1.5 or diameter, d=h/1.5
- Velocity=distance (height) /time

Example: settling of dextran beads

Data: $d = 150\ \mu m = 0.015$ cm; $\mu = 1.1$ cP $= 0.011$ g/cm-s; $\rho_s = 1.02$ g/cm³; $\rho = 1.00$ g/cm³; $g = 980$ cm/s²

Hint: $v_g = \frac{d^2}{18\mu}(\rho_s - \rho)g$

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Data: d = 150 μm = 0.015 cm; μ = 1.1 cP = 0.011 g/cm-s; ρ_s = 1.02 g/cm³; ρ = 1.00 g/cm³; g = 980 cm/s²

(a) Estimate the settling time by assuming that these beads quickly reach their maximum terminal velocity.

Solution:

$$v_g = \frac{d^2}{18\mu}(\rho_s - \rho)g \quad \square \quad v_g = 0.022 \text{ cm/s}$$

Check: $N_{Re} = \frac{\rho v d}{\mu} = \frac{1 \times 0.022 \times 0.015}{0.011} = 0.03 < 1$

Liquid volume, $V = \left(\frac{\pi d_t^2}{4}\right) \times h = \frac{\pi}{4}\left(\frac{h}{1.5}\right)^2 h = 50 \text{ L}$

$\square$ h = 52.3 cm $\square$ **Settling time** $= \frac{h}{v_g} = \frac{52.3 \text{ cm}}{0.22 \text{ cm/s}} = 2379 \text{ s}$

(To be continued)

- Check the validity of using the Free settling velocity or terminal velocity or stoke's law settling
- Volume of stirred tank (cylinder)=$\pi r^2 h$
- r=d/2
- Height/diameter= 1.5 or diameter, d=h/1.5
- Velocity=distance (height) /time

50L=50000cm3

ISOPYCNIC SEDIMENTATION

- ☞ To capture particles in a solution having density gradient.
- ☞ Application: determining the density of the solute or suspended particle.

* There are three methods for establishing conditions for isopycnic sedimentation:

(1) Layer solutions of decreasing density, starting at the bottom of the tube.

(2) Centrifuge the solution containing a density-forming solute (such as CsCl) at extremely high speed.

(3) Use the gradient mixing method.

- Isopycnic- separating particles that differ in density but not in their size.
- **Isopycnic sedimentation** (also called **density-gradient sedimentation**) is a separation process in which particles move through a **density gradient** until they reach a position where the **density of the surrounding fluid equals their own density**.
- At that point:

$$\rho_{particle} = \rho_{medium}$$

- and the **net gravitational (or centrifugal) force becomes zero,** so the particle **stops moving**. That equilibrium position is called the **isopycnic point** (iso = same, pycno = density).

DIFFERENTIAL SETTLING (or CLASSIFICATION)

☞ **Separation of solid particles into several size fractions based upon the settling velocities in a medium.**

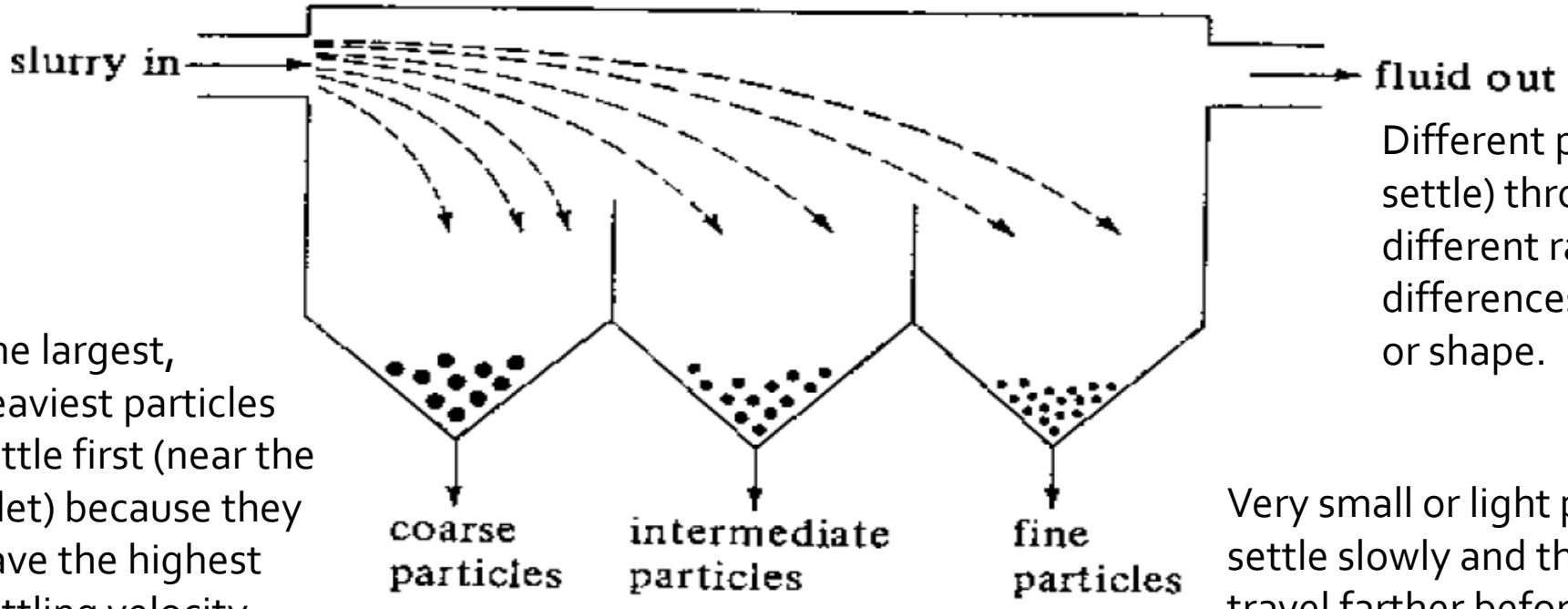


FIGURE 14.3-6. *Simple gravity settling classifier.*

Different particles fall (or settle) through the fluid at different rates because of differences in size, density, or shape.

The largest, heaviest particles settle first (near the inlet) because they have the highest settling velocity.

Very small or light particles settle slowly and therefore travel farther before settling, or may even exit with the overflow fluid.

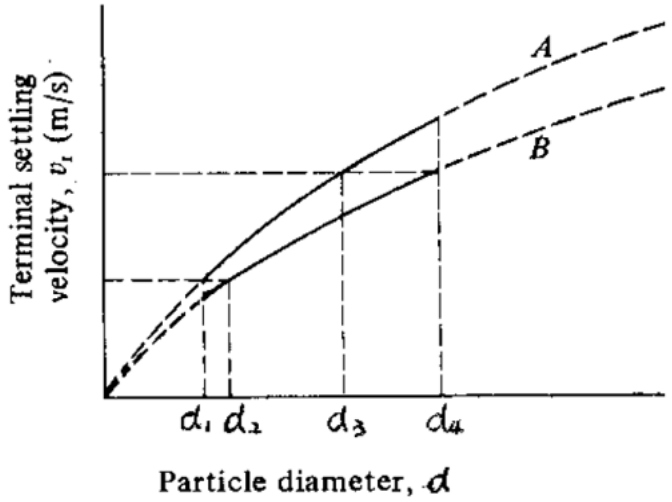
If the light and heavy materials both have a range of particle sizes, the smaller, heavy particles settle at the same terminal velocity as the larger, light particles.

The terminal settling velocities of components A and B are:

$$v_{gA} = \sqrt{\frac{4(\rho_{sA} - \rho)gd_A}{3\rho C_{DA}}} \quad \text{and} \quad v_{gB} = \sqrt{\frac{4(\rho_{sB} - \rho)gd_B}{3\rho C_{DB}}}$$

For particles of equal settling velocities, $v_{gA} = v_{gB}$.

$$\frac{(\rho_{sA} - \rho)d_A}{C_{DA}} = \frac{(\rho_{sB} - \rho)d_B}{C_{DB}} \quad \text{or} \quad \frac{d_A}{d_B} = \frac{\rho_{sB} - \rho}{\rho_{sA} - \rho} \times \frac{C_{DA}}{C_{DB}}$$



A- heavy particles
B-lighter particles
Both A and B have a range of particle size, d1, d2, d3 and d4
The smaller particle size (d3) of heavy particle, settle at the same terminal velocity as the larger particle size (d4) of light particles

- Can determine the diameter of the other particles if some diameters are given

$$\frac{d_A}{d_B} = \frac{\rho_{sB} - \rho}{\rho_{sA} - \rho} \times \frac{C_{DA}}{C_{DB}}$$

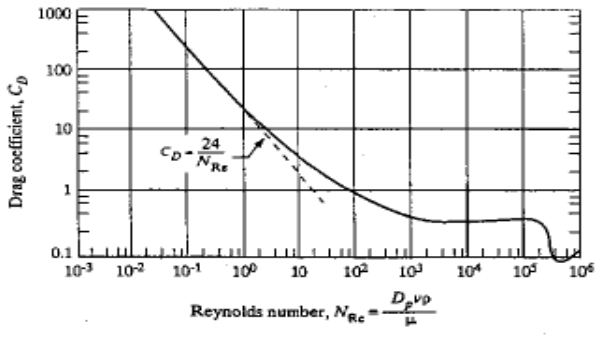


FIGURE 14.3-1. *Drag coefficient for a rigid sphere.*

In the turbulent Newton's law region, C_D is constant.

$$\frac{d_A}{d_B} = \frac{\rho_{sB} - \rho}{\rho_{sA} - \rho} \times \frac{C_{DA}}{C_{DB}} \quad \square \quad \frac{d_A}{d_B} = \frac{\rho_{sB} - \rho}{\rho_{sA} - \rho}$$

For laminar Stokes' law settling,

$$C_{DA} = \frac{24\mu}{\rho v_{gA} d_A} \quad \text{and} \quad C_{DB} = \frac{24\mu}{\rho v_{gB} d_B}$$

$$\frac{d_A}{d_B} = \frac{\rho_{sB} - \rho}{\rho_{sA} - \rho} \times \frac{C_{DA}}{C_{DB}} = \frac{\rho_{sB} - \rho}{\rho_{sA} - \rho} \times \frac{d_B}{d_A} \quad \square \quad \frac{d_A}{d_B} = \left(\frac{\rho_{sB} - \rho}{\rho_{sA} - \rho} \right)^{0.5}$$

$$\frac{d_A}{d_B} = \frac{\rho_{sB} - \rho}{\rho_{sA} - \rho} \times \frac{C_{DA}}{C_{DB}}$$

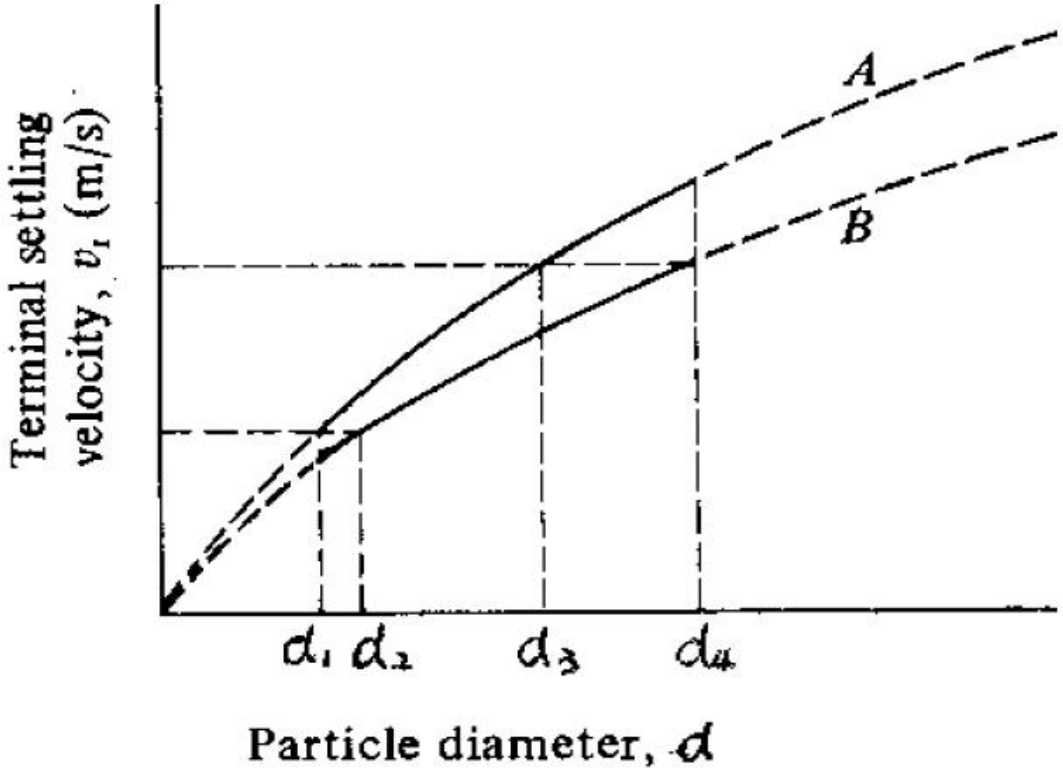
In the turbulent Newton's law region, C_D is constant, $\frac{d_A}{d_B} = \frac{\rho_{sB} - \rho}{\rho_{sA} - \rho}$

For laminar Stokes' law settling, $\frac{d_A}{d_B} = \left(\frac{\rho_{sB} - \rho}{\rho_{sA} - \rho} \right)^{0.5}$

For transition flow between laminar and turbulent flow,

$$\frac{d_A}{d_B} = \left(\frac{\rho_{sB} - \rho}{\rho_{sA} - \rho} \right)^{n} \quad \text{where} \quad \frac{1}{2} < n < 1$$

- **Settling a mixture of particles of materials A (the heavier) and B (the lighter) with a size range of d_1 to d_4 for both types of material:**

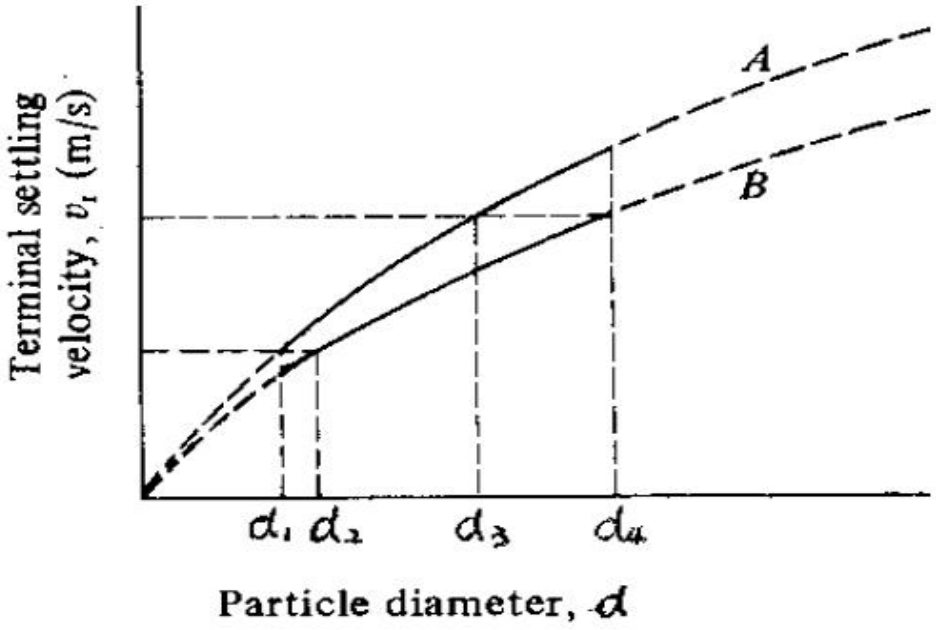


* **Size range d_{A3} to d_{A4}: pure fraction of A**

☞ **No B particles settle as fast as the A particles in this size range.**

* **Size range d_{B1} to d_{B2}: pure fraction of B**

☞ **No particles of A settle as slowly.**



* **Size range of A particles from d_{A1} to d_{A3} and size range of B particles from d_{B2} to d_{B4}: form a mixed fraction of A and B**

* **Increasing the density ρ of the medium.**

☞ **The spread between d_A and d_B is increased.**

[Example] A mixture of silica and galena (方鉛礦 ; PbS) solid particles having a size range of 5.21×10^{-6} m to 2.50×10^{-5} m is to be separated by hydraulic classification using free settling conditions in water at 20°C. The specific gravity of silica is 2.65 and that of galena is 7.5. Calculate the size range of the various fractions obtained in the settling. The water viscosity at 20°C is 1.005×10^{-3} Pa-s.

Solution: A particles: galena; B particles: silica

Assuming Stokes' law settling, $v_g = \dfrac{d^2}{18\mu}(\rho_s - \rho)g$

☞ Check the validity of the Stokes' law region.

(To be continued)

Solution (cont'd):

$$SG = \frac{\rho_{object}}{\rho_{reference}}$$

where

- ρ_{object} = density of the material (g/cm^3 or kg/m^3)
- $\rho_{reference}$ = density of reference (usually water = 1 g/cm^3 or 1000 kg/m^3 at 4 °C)

$$v_g = \frac{d^2}{18\mu}(\rho_s - \rho)g$$

For the largest particle and the biggest density,

$$d_A = 2.50 \times 10^{-5}\ \text{m} \quad \text{and} \quad \rho_{sA} = 7.5\ \text{g/cm}^3 = 7500\ \text{kg/m}^3$$

$$v_{gA} = \frac{(2.50\times10^{-5})^2}{18(1.005\times10^{-3})}(7500-1000)(9.8) = 2.20\times10^{-3}\ \text{m/s}$$

Check : $N_{Re} = \frac{\rho v d}{\mu} = \frac{1000(2.20\times10^{-3})(2.50\times10^{-5})}{1.005\times10^{-3}} = 0.0547 < 1$

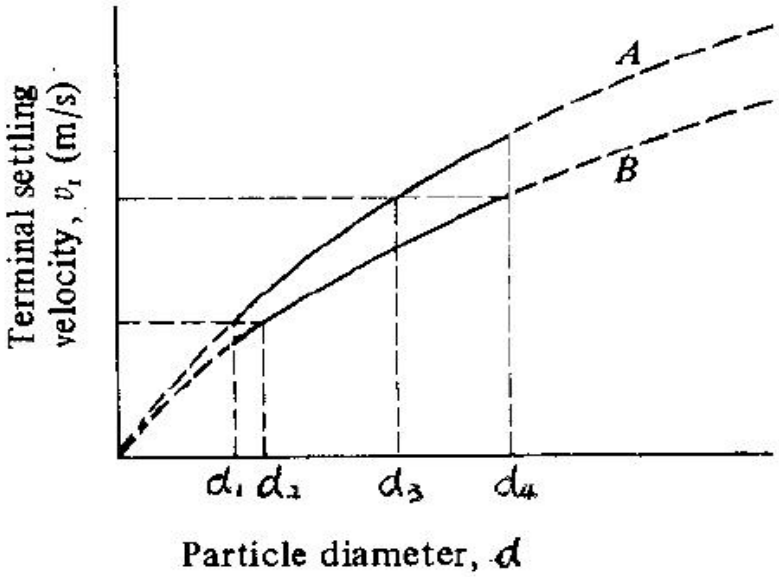
☞ **O.K. with the Stokes' law region.**

(To be continued)

A particles: galena; B particles: silica

solid particles having a size range of 5.21×10^{-6} m to 2.50×10^{-5} m is to be separated by hydraulic classification using free settling conditions in water at 20°C. The specific gravity of silica is 2.65 and that of galena is 7.5. Calcul

Solution (cont'd):



For particles of equal settling velocities,

$$\frac{d_A}{d_B} = \left(\frac{\rho_{sB} - \rho}{\rho_{sA} - \rho}\right)^{0.5}$$

$$\frac{d_{A3}}{d_{B4}} = \left(\frac{\rho_{sB} - \rho}{\rho_{sA} - \rho}\right)^{0.5}$$

$$\Rightarrow \quad \frac{d_{A3}}{2.50 \times 10^{-5}} = \left(\frac{2.65 - 1}{7.5 - 1}\right)^{0.5}$$

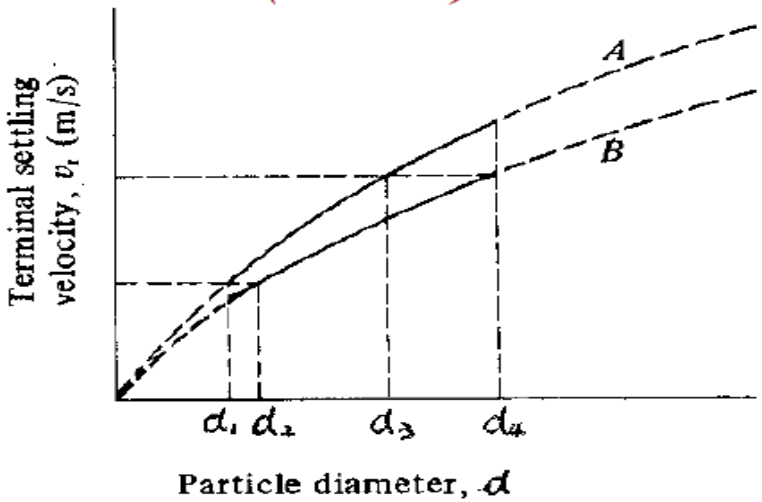
$$\Rightarrow \quad d_{A3} = 1.260 \times 10^{-5} \text{ m}$$

The size range of pure A (galena) is:

$$d_{A3} = 1.260 \times 10^{-5} \text{ m} \quad \text{to} \quad d_{A4} = 2.50 \times 10^{-5} \text{ m}$$

(To be continued)

Solution (cont'd):



$$\frac{d_{A1}}{d_{B2}} = \left(\frac{\rho_{sB} - \rho}{\rho_{sA} - \rho}\right)^{0.5}$$

$$\frac{5.21 \times 10^{-6}}{d_{B2}} = \left(\frac{2.65 - 1}{7.5 - 1}\right)^{0.5}$$

$$d_{B2} = 1.033 \times 10^{-5} \text{ m}$$

The size range of pure B (silica) is:

$$d_{B1} = 5.21 \times 10^{-6} \text{ m to } d_{B2} = 1.033 \times 10^{-5} \text{ m}$$

The mixed-fraction size range is:

$$d_{A1} = 5.21 \times 10^{-6} \text{ m to } d_{A3} = 1.260 \times 10^{-5} \text{ m}$$

$$d_{B2} = 1.033 \times 10^{-5} \text{ m to } d_{B4} = 2.50 \times 10^{-5} \text{ m}$$

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